CJC-1295 NODAC vs CJC-1295 DAC vs Tesamorelin

CJC-1295 NODAC, CJC-1295 DAC, and Tesamorelin are studied for GH/IGF-1 augmentation and body-composition effects, with visceral-fat reduction demonstrated most clearly for tesamorelin. [3,4,7–9,12] All three are GHRH-receptor agonists that stimulate pituitary GH release and downstream IGF-1 signaling. [1–6] NODAC lacks the albumin-binding appendage and plausibly concentrates stimulation near administration and for a very short duration; DAC prolongs exposure through albumin binding primarily increasing basal GH secretion. [2–4,26] Tesamorelin increases basal and pulsatile GH secretion despite brief plasma exposure. [5,6] Adding ipamorelin offers plausible complementary stimulation through the ghrelin receptor. [16–18]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Biological Foundation

GHRH is released from the hypothalamus and acts on somatotroph cells in the anterior pituitary. GHRH-receptor activation engages Gs/adenylyl cyclase, raises cyclic AMP, and promotes GH synthesis and release. Somatostatin opposes release, while GH and IGF-1 participate in feedback control. [1,6] All three compounds mimic the GHRH signal and therefore depend on functioning pituitary somatotrophs. [2–6,26]

Released GH binds GH receptors in liver, adipose tissue, muscle, bone, and other tissues. Some actions are direct, including stimulation of lipolysis; others involve hepatic and local IGF-1. [1,21,26] GH-receptor signaling includes JAK–STAT pathways, and IGF-1 activates pathways involved in growth, survival, and tissue remodeling. [23,26] Raising GH or IGF-1 is a biochemical effect creating a cascade of downstream effects.

Drug half-life, duration of receptor stimulation, GH pulse pattern, and duration of IGF-1 elevation are distinct biomarkers that vary in the three compounds. A peptide can disappear from plasma quickly yet influence downstream hormones for longer. Conversely, a long-acting GHRH stimulus can coexist with pulsatile GH secretion because endogenous secretion remains regulated. [3,4,6]

CJC-1295 NODAC

Identity and structure: CJC-1295 NODAC, GRF(1–29), is the 29-amino-acid, tetrasubstituted GHRH fragment without the albumin-binding DAC appendage. The substitutions increase stability relative to native GRF, while retaining the GHRH-receptor pharmacologic rationale. (The CJC-1295 used in the principal published human trials included DAC; the names are not interchangeable evidence labels.) [2–4,26]

Operation in the body: After administration, the no-DAC analog is expected to provide a relatively brief GHRH-receptor stimulus, followed by peptide clearance. A precise human elimination half-life for the exact modified GRF(1–29) compound remains uncertain. A related analog containing only the D-Ala^2 substitution had a measured half-life of 6.7 ± 0.5 minutes, versus 4.3 ± 1.4 minutes for unmodified GRF(1–29); these values are suggestive of a very short-half life for NODAC, but cannot be assigned directly. Enhanced pituitary GH release is likely; downstream IGF-1 elevation is plausible. [1,2,24,26,29]

Distinctive functional expectation: Stimulation is plausibly confined to the administration window. Intermittent stimulation and timing-dependent amplification of an endogenous GH burst are mechanistic expectations. [1,2,26]

CJC-1295 DAC

Identity and structure: DAC-form CJC-1295 contains a modified GHRH(1–29) backbone plus a reactive C-terminal lysine derivative. The drug-affinity complex binds covalently to the free thiol of albumin, creating a circulating reservoir that prolongs exposure while preserving GHRH-receptor activity. Its mechanism was characterized in rat and cell experiments. [2]

Operation in the body: In healthy-adult trials lasting 28 and 49 days, the estimated half-life was 5.8–8.1 days. After a single SC dose, mean GH rose approximately 2–10-fold for at least 6 days, and mean IGF-1 rose approximately 1.5–3-fold for 9–11 days. Repeated doses produced accumulation and sustained IGF-1 elevation. [3]

Distinctive functional effect: A human overnight-sampling study found preserved GH pulsatility with a substantial rise in trough GH. Pulse frequency and magnitude were not significantly changed. DAC produces a more sustained background stimulus, but does not eliminate pulses. [3,4]

Tesamorelin

Identity and structure: Tesamorelin is a 44-amino-acid human GHRH analog with an N-terminal hexenoyl group. It activates the GHRH receptor without a DAC albumin-binding appendage. [5]

Operation in the body: In small trials tesamorelin 2 mg SC daily for 2 weeks increased mean overnight GH from 0.7 to 1.2 µg/L and IGF-1 from 148 to 329 µg/L. Basal GH secretion and GH pulse area both increased. [6]

Distinctive functional effect: Tesamorelin provides brief peptide exposure resulting in longer lasting GH stimulation in clinical studies. Mean elimination half-life of the original tesamorelin acetate formulation was 26 – 38 minutes after 14 consecutive days of subcutaneous administration. [5–9]

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It demonstrated reduction of visceral abdominal fat, the fat around internal organs, distinct from subcutaneous fat under the skin; tesamorelin is characterized as altering fat distribution rather than as a general weight-loss agent. [7–9,12]

Comparison at a glance

Feature	CJC-1295 NODAC	CJC-1295 DAC	Tesamorelin
Structure	Modified 29-aa GRF; no albumin linker. [2,26]	Modified GRF plus albumin-binding appendage. [2]	44-aa GHRH analog; N-terminal modification. [5]
Duration	Relatively brief stimulus; exact human PK uncertain. [2,26]	Human half-life 5.8–8.1 days. [3]	Brief exposure; repeat daily stimulation. [5,6]
GH pattern	Likely GH enhancement; timing-dependent mechanistic expectation. [2,26]	Pulsatility preserved; trough GH rises. [4]	Basal secretion and pulse area increase. [6]
IGF-1	Plausible increase. [1,2,26]	Sustained human increase. [3,4]	148 to 329 µg/L after 2 weeks in 13 men. [6]
Fat outcomes	Plausible GH-mediated lipolysis. [1,22,26]	Plausible GH-mediated lipolysis. [3,4,26]	Visceral-fat reduction in defined human populations. [7,8,12]
Distinctive strength	Brief-stimulation rationale. [2,26]	Long-lasting hormone response. [3,4]	Direct fat and metabolic outcome trials. [7–12]
Ipamorelin pairing	Plausible overlap of two brief inputs. [16–18]	Plausible transient input on sustained background. [3,16–18]	Plausible complementary receptor stimulation. [6,16–18]

Interaction with natural pulsatile GH secretion

Natural GH secretion consists of bursts separated by periods of lower secretion. Their timing and size reflect GHRH, somatostatin, ghrelin signaling, and feedback from GH and IGF-1. Pulse frequency, pulse size, and between-pulse secretion are distinct features. [1,4,6]

CJC-1295 NODAC: A brief GHRH-receptor stimulus plausibly enhances an endogenous burst or evokes an additional burst, depending on administration timing, somatostatin inhibition, and pituitary responsiveness. Synchronization with a natural nighttime pulse is an unproven mechanistic hypothesis. [1,2,26]

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CJC-1295 DAC: Sustained stimulation can coexist with natural GH pulses. In healthy men assessed one week after a single dose, pulse frequency and magnitude were unchanged, while trough GH increased approximately 7.5-fold, mean GH increased 46%, and IGF-1 increased 45%. DAC therefore raised the background between pulses without eliminating pulsatility in that study. [4]

Tesamorelin acetate, original formulation: In 13 men receiving 2 mg daily for two weeks, tesamorelin increased GH pulse area and basal secretion. Evidence is mixed on increasing pulse frequency. Trial results show Tesamorelin can effect endogenous secretion well beyond the peptide's plasma half-life. [6]

Comparison: DAC's clearest measured pulsatility effect was increased between-pulse GH; tesamorelin increased both pulse area and basal secretion. [4,6] NODAC's comparatively brief stimulus provides a plausible rationale for more localized stimulation. [2,26] The DAC and tesamorelin measurements came from different studies, doses, and sampling schedules; their numerical responses do not establish relative potency. [3,4,6]

Expected functional effects

- **GH release and IGF-1:** Human studies establish increases with DAC and tesamorelin. NODAC likely enhances GH release, with plausible downstream IGF-1 elevation. There is no reliable dose-equivalent ranking of the three from these separate studies. [2–6,26]
- **Time course:** NODAC and tesamorelin support a brief-stimulation model; DAC maintains receptor stimulation over days. Tesamorelin has measured human short-duration PK, whereas the exact human PK of NODAC remains insufficiently characterized. [2–6,24,26,30]
- **Pulse pattern:** DAC preserved pulsatility while raising trough GH, and tesamorelin increased basal and pulsatile secretion. A short-lived GHRH analog does not, by itself, establish a safer or more beneficial pulse pattern. [4,6]
- **Pituitary reserve and feedback:** All rely on endogenous GH-release capacity. Age-related decline, obesity-associated suppression, and diagnosed GH deficiency are distinct states. [1,6,12] Feedback plausibly moderates stimulation. [1,26]
- **Glucose handling:** Higher GH can impair insulin sensitivity. [22] Short tesamorelin studies sometimes preserved average insulin sensitivity, and selected longer trials did not worsen average glucose measures. [6,10–12] Nevertheless, diabetes defined by HbA1c ≥6.5% developed in 5% of tesamorelin-treated versus 1% of placebo-treated participants over 26 weeks in the labeling dataset. [30]

Expected benefits and their differences

The clearest benefit difference is evidentiary: tesamorelin has direct trials measuring fat and metabolic outcomes. [7–12] DAC has convincing hormone-response evidence, while many clinical benefits proposed for DAC or NODAC remain plausible extrapolations. [3,4,26] Broader healthspan applications are speculative.

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- **Visceral-fat reduction:** Tesamorelin has established efficacy for visceral fat reduction. [7-8,12] NODAC and DAC have plausible lipolytic rationales through GH signaling. [1,3,4,26]
- **Liver-fat reduction:** Tesamorelin reduces liver fat. [10] Broader liver protection remains plausible. [11] Corresponding NODAC or DAC benefits are mechanistic extrapolations through the shared GH/IGF-1 axis. [1,2,26]
- **Triglycerides and risk markers:** Tesamorelin improved triglycerides and other lipid biomarkers in trials. [7,9] These are distinct from heart attacks or stroke. [12]
- **Lean mass and muscle growth:** GH-related studies provide a plausible rationale for changes in lean mass across the three. Lean mass includes water and noncontractile tissue and is distinct from muscle strength or muscle-fiber growth. [22,26]
- **Connective tissue and recovery:** Collagen turnover and tissue-repair signaling provide plausible benefits through the shared GH/IGF-1 axis. The human collagen findings used administered GH. [21] Accelerated tendon healing, joint-pain relief, and reduced exercise soreness from a specific analog remain mechanistic extrapolations. [21,26]
- **Sleep and energy:** GHRH biology provides a plausible sleep-related rationale. GHRH promoted sleep in young men under specific timing conditions and impaired sleep in young women. [27,28]
- **Cognition:** Tesamorelin has mixed human findings.. [13-15] Cognitive improvement is plausible and population-dependent, with more indirect extrapolation to NODAC or DAC.
- **Skin quality and skeletal support:** Shared collagen and growth signaling provide plausible support for skin and bone remodeling. Cosmetic improvement, increased bone strength, and fracture prevention remain endpoint-specific extrapolations rather than consequences established by a higher IGF-1 measurement alone. [1,21,22,26]

Combining each with ipamorelin

Ipamorelin is a ghrelin-receptor (GHSR-1a) agonist, providing a different input to the GH-release system. [1,16,26] Human IV studies demonstrated an episode of GH release. [17] Swine experiments showed relatively selective GH stimulation, with ACTH/cortisol responses comparable to GHRH and no measured prolactin increase. Comparable human endocrine selectivity is plausible. [16]

A human trial of GHRP-6 combined with GHRH demonstrated synergistic GH release. It provides a plausible rationale for adding ipamorelin to any of the three GHRH analogs. [18] Additional visceral-fat loss, recovery, or dose sparing from those exact pairings are speculative downstream effects. [1,16–18,26]

- **NODAC plus ipamorelin:** Two relatively brief stimuli can plausibly overlap near administration, enhancing the GH response through complementary receptors. This provides a mechanistic argument for a temporally paired regimen. [1,2,16–18] Fasting or bedtime administration is an empirical practice. [26]
- **DAC plus ipamorelin:** Albumin-bound DAC provides a sustained GHRH background, while each ipamorelin administration can plausibly add a more transient secretagogue input. Precise simultaneous dosing is therefore less central to the overlap rationale than with two

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brief agents. The pair could increase cumulative GH/IGF-1 exposure while retaining DAC's sustained background stimulus. [2–4,16–18]

- **Tesamorelin plus ipamorelin:** The brief GHRH stimulus and ghrelin-receptor stimulus plausibly complement one another, similar to NODAC pairing. [6,16–18] Tesamorelin's demonstrated clinical benefits concern monotherapy; greater benefit or dose sparing with ipamorelin is speculative. [7–12,26]
- **Fat-loss qualification:** Ipamorelin increased food intake and adiposity in a mouse study through effects partly independent of GH. This does not establish human weight gain, but cautions against assuming that a larger GH response means greater fat loss. The direction of additional fat-loss benefit from any pairing remains uncertain. [19]
- **Additional skeletal rationale:** Ipamorelin increased bone mineral content and bone dimensions in adult female rats over 12 weeks, without increasing volumetric bone mineral density. The increase in total mineral content did not persist after correction for body weight. These findings support plausible skeletal effects in humans and from the combinations; fracture prevention remains speculative. [20]

Dosing patterns and persistence of effect

The following are studied experimental regimens, rather than individualized prescriptions. DAC studies measured hormone responses; tesamorelin trials generally used the original formulation for clinical outcomes. [3,5,7–12,24]

- **CJC-1295 DAC: 1.5 - 2 mg SC weekly for three injections, or 2 - 48 mg SC every 14 days for two injections**, based on a 150 lb subject.

- **Tesamorelin: 2 mg SC daily, generally for 26–52 weeks in visceral-fat trials. [5,7–9]**

NODAC and ipamorelin community regimens often are in the range of 100 – 250 mcg, including fasted or bedtime timing. Their optimal dose, timing, cycling, and combination ratio are empirical. A higher nominal dose cannot be assumed to improve benefits, and milligram-for-milligram comparisons between structurally different agents are misleading. [17,26]

Safety and monitoring differences

Shared GH-axis concerns include fluid retention, musculoskeletal discomfort, and dysglycemia. [22,30] DAC's persistence makes exposure slower to reverse after stopping. [3]

Tesamorelin labeling contraindicates active malignancy, pregnancy, disrupted hypothalamic-pituitary function, and hypersensitivity; glucose and IGF-1 monitoring are recommended. [30] GH/IGF-1 signaling plausibly supports growth of responsive tumors. In contrast, the GH-replacement consensus found no association with primary cancer recurrence in the reviewed replacement populations. [23]

Hepatic outcomes: Tesamorelin's 12-month trial showed reduced liver fat and less fibrosis progression. Continued hepatic tolerability is plausible under comparable monitoring; cirrhosis

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was excluded. [11] DAC trials found no consistent liver-laboratory changes during their short observation periods. [24]

Fluid retention and musculoskeletal outcomes: Edema, arthralgia, and carpal tunnel symptoms are recognized tesamorelin effects and are often transient or resolve after withdrawal. [30] In patients stable on treatment, continued tolerability of these common effects is plausible from the 52-week extension experience. [9] That inference is conditional on dose, symptoms, and ongoing monitoring.

Endocrine and metabolic outcomes: Dysglycemia remains a monitored risk even when mean glucose is stable. [9–12,30] At 52 weeks, 23% of participants remaining on tesamorelin had IGF-1 above 3 standard deviation scores. Prolonged supraphysiological IGF-1 exposure should therefore be assessed separately from the presence of GH pulses. [30]

Cardiovascular and neurological outcomes: DAC studies reported transient hypotension and increased heart rate. FDA also summarized a myocardial-infarction death in an unpublished trial; available information did not establish drug causation. These observations do not support a confident low-risk extrapolation for repeated long-term DAC exposure. [24] Improvements in tesamorelin vascular surrogates are distinct from evidence about cardiovascular events. [12] Mixed cognition studies likewise cannot establish broad neurological protection. [13–15]

Immunological outcomes: Tesamorelin hypersensitivity occurred in 4% of treated participants; injection-site reactions occurred in 25% versus 14% with placebo during 26 weeks. [30]

Proliferative and reproductive outcomes: Reassuring experience with physiological GH replacement in selected cancer survivors supports a context-specific clinical inference, while secretagogue combinations and persistent high IGF-1 require separate assessment. [23] Rare, delayed malignancy outcomes remain inadequately characterized for these regimens. Tesamorelin is contraindicated during pregnancy; reproductive safety cannot be extrapolated from adult metabolic trials. [30]

Ipamorelin combination safety: FDA's review of an IV postoperative study noted hypokalemia, hyperglycemia, and two deaths in the ipamorelin arm, with uncertain causal attribution. The surgical setting and IV exposure differ from proposed SC combinations. [25]

Conclusions

NODAC is best distinguished by a brief-stimulation design and predominantly indirect clinical-benefit evidence. DAC is best distinguished by measured prolonged exposure and sustained GH/IGF-1 elevation with preserved pulsatility. Tesamorelin is best distinguished by direct human evidence for visceral-fat reduction, selected liver and metabolic outcomes, and a defined formulation-specific clinical use. The three share a receptor pathway but are not evidence-equivalent or interchangeable. [2–12,26]

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For each, ipamorelin offers plausible complementary GH-release synergy. [16–18] Muscle growth, recovery, sleep, cognition, and healthspan applications depend on the specific outcome and population; their mechanistic plausibility is distinct from the magnitude of the hormone response. [13–23,26–28]

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